

# Event-Triggered Predictive Control for Networked Nonlinear Systems With Imperfect Premise Matching

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**Abstract**—This paper investigates the event-triggered predictive control problem for networked nonlinear systems with imperfect premise matching. First, a model of networked nonlinear system is well constructed, which has integrated the event-triggered communication scheme (ETCS) and the predictive control together, in which, an ETCS is introduced to alleviate the communication burden by reducing the number of transmitted packets; and a fuzzy predictive controller is designed to predict future states and control signals between two successfully transmitted instants. Therefore, the data dropout induced by the networks can be actively compensated. Second, by using a common Lyapunov theory, a stability criterion and two stabilization criteria are deduced to ensure the asymptotical stability of the studied system and find the controller gains, respectively. Different from the traditional parallel distributed compensation method, the synchronous premise variables between the T-S fuzzy system and the fuzzy event-triggered predictive controller (FETPC) are no longer needed again. Since the imperfect premise matching condition is well considered in the derivation of the main results, the design flexibility and low cost of the FETPC implementation can be expected. Finally, the validity of the method proposed in this paper is demonstrated by a nonlinear mass-spring example.

**Index Terms**—Event-triggering communication, imperfect premise matching, predictive control, T-S fuzzy systems.

## I. INTRODUCTION

CONTROL systems wherein the shared communication networks connect various components those are physically decentralized are called networked control systems [1]. They have been widely used in many engineering applications, such as underwater acoustics, arrays of microactuators, vehicles, and crafts, where signals transmit through bandwidth-limited communication networks [2], [3]. Motivated by their widespread applications, networked nonlinear systems have attracted more attention during the past decades. See, for example, [4]–[6].

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In terms of nonlinear systems, the T-S fuzzy models can be qualified to express certain systems with nonlinear dynamics, which describe the nonlinear systems through a group of local linear subsystems blended by several IF-THEN rules [7]–[10]. Under consideration of the sensor and nonaffine faults, Yang *et al.* presented novel fault detection and isolation method and fault-tolerant tracking control method for nonlinear systems [11], [12] with point-to-point connection. Taking the introduced communication networks into account, the characteristics of communication networks, such as packet dropout and time delay occurring in the network transmissions, may lead to the degradation of the control system performance. With the purpose of compensating for the data dropout or time delay, the networked predictive control methods have been proposed in [18]–[20] and [23]. Nevertheless, in these previous predictive control schemes, the sensors are time triggered, implying that all sampled data must to be transmitted over the bandwidth-limited communication networks. As an improvement, Yue *et al.* [24] proposed an event-triggered predictive control (ETPC) method to compensate for the negative effect of the time delay, in which whether or not the sampled data should be transmitted depends not only on the current-sampled state but also on the error between the latest transmitted states and current-sampled states. Compared with the previous communication scheme periodically driven by the clock, event-triggered communication scheme (ETCS) has a novel superiority to keep the expected control performance with much less bandwidth occupancy [13]. However, since there are communication networks between the sensor and the controller in the studied T-S fuzzy system, the assumption of identical premises between the T-S fuzzy system and the fuzzy predictive controller is not kept again as the widely used traditional parallel distributed compensation (PDC) technology [15]–[17].

To tackle the above-mentioned asynchronous premise problem, there are some nice results on designing of networked fuzzy controller [21], [25], [26]. For example, assuming that the distribution of the membership function is known *a priori*, Peng *et al.* [25] reconstructed the grades of membership at the controller by storing the grades of membership of the fuzzy plant to a buffer and using a time-stamping technique, which guarantees that the membership of the fuzzy plant and the controller has the same time scale. Without depending on any information of the premises between the T-S fuzzy system and the fuzzy controller, Lam *et al.* [26], [31] introduced a zero sum item to connect the different time-scale membership functions for the main results. By use of an observer to estimate the unmeasurable premise

variable, Li *et al.* [21] and Peng *et al.* [22] presented novel approaches to designing controller for networked T-S fuzzy systems with unmeasurable premise variables and imperfect premise matching, respectively. Compared with the traditional PDC method, the methods in [21] and [22] facilitate the design flexibility, and reduce both the difficulty of practical application and the cost of controller implementation. However, how to deal with the data dropout in the communication is not mentioned in the above work; this issue also motivates this paper.

The goal of this paper is to design a fuzzy event-triggered predictive controller (FETPC) for networked T-S fuzzy systems, which has the different premises as those of the T-S fuzzy model and is able to compensate for the negative effect induced by the communication networks. First, a new networked T-S fuzzy system model is provided, including the ETCS to alleviate the communication burden, and the predictive control scheme to tackle the data-dropout problem in the transmissions. Then, by using the common Lypunov theory and imperfect premise matching method, three theorems are derived to judge the stability and stabilization of the studied system, and the FETPC gains and the event-triggering parameters can be solved out based on the given stabilization criterion. The major contributions of this paper can be generalized as follows.

- 1) A novel closed-loop model is constructed for the networked T-S fuzzy system unifying the event-triggering transmission and the predictive control in a framework, which actively compensates for the negative effect of networks with less network bandwidth occupancy. Compared with those existing results in [18]–[20] and [23], the number of transmitted sampled states and the size of predictive control packets are greatly reduced to alleviate the communication burden.
- 2) An approach to design an FETPC is provided for networked T-S fuzzy systems with imperfect premise matching. Different from the conventional PDC method [4], it is not required that the controller has the same premise variables as the studied T-S fuzzy system. Different from networked non-PDC method [25], the distribution of premise membership is not necessary to known *a priori*.

*Notation:* Throughout the paper,  $\mathbb{R}^n$  and  $\mathbb{R}^{n \times m}$  stand for the  $n$ -dimensional Euclidean space and the set of  $n \times m$  real matrices. The superscript “ $T$ ” means the transposition of the matrices. For matrix  $X \in \mathbb{R}^{n \times n}$ , the notation  $X > 0$  (respectively,  $X \geq 0$ ) denotes that  $X$  is a real symmetric positive-definite (positive semidefinite) matrix. In symmetric block matrices,  $*$  is utilized as ellipsis for terms induced by symmetric, and  $\text{diag}\{\dots\}$  represents the block-diagonal matrix.

## II. SYSTEM DESCRIPTION AND PRELIMINARIES

### A. System Description

Consider a networked T-S fuzzy system, as depicted in Fig. 1, where there are communication networks between the sensor and the controller, and the controller and the actuator are directly point-to-point connected. The controlled plant can be described by a model with  $v_{th}$  rule of the model as follows:

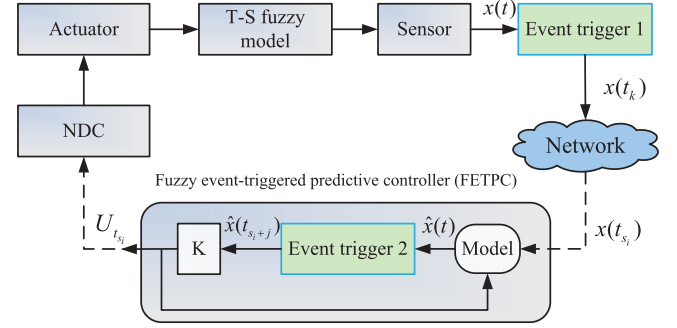


Fig. 1. Diagram of a networked T-S fuzzy control system.

Plant rule  $v$ : IF  $f_1(x(t))$  is  $W_1^v$  and  $\dots$  and  $f_q(x(t))$  is  $W_q^v$ , THEN

$$x(t+1) = A_v x(t) + B_v u(t) \quad (1)$$

where  $W_b^v$  ( $v = 1, 2, \dots, \bar{r}$ ;  $b = 1, 2, \dots, q$ ) denote the fuzzy sets;  $f_b(x(t))$  denote the premise variables;  $\bar{r}$  and  $q$  are the number of fuzzy IF-THEN rules and the premise variables, respectively;  $x(t) \in \mathbb{R}^n$  and  $u(t) \in \mathbb{R}^m$  are the state vector and the control input vector; and  $A_v$  and  $B_v$  are constant system matrices with appropriate dimensions.  $x(t_0) = x_0$  is the initial state. For simplicity,  $f_b(x)$  represent  $f_b(x(t))$ , and  $f(x) = [f_1(x), \dots, f_q(x)]^T$ .

Applying product inference, a standard singleton fuzzifier, and a center-average defuzzifier [29], the initial system (1) can be inferred as

$$x(t+1) = \sum_{v=1}^{\bar{r}} \mu_v(f(x)) [A_v x(t) + B_v u(t)] \quad (2)$$

where

$$\mu_v(f(x)) = \frac{\alpha_v(f(x))}{\sum_{v=1}^{\bar{r}} \alpha_v(f(x))}, \alpha_v(f(x)) = \prod_{b=1}^q W_b^v(f_b(x)) \quad (3)$$

and  $W_b^v(f_b(x))$  is the membership value of  $f_b(x)$  in  $W_b^v$ . Therefore, we can see from (3) that for all  $v \in \{1, \dots, \bar{r}\}$

$$\mu_v(f(x)) \geq 0, \sum_{v=1}^{\bar{r}} \mu_v(f(x)) = 1. \quad (4)$$

The T-S fuzzy system (2) is controlled by an ETPC scheme. The scheme proposed in this paper includes event trigger 1 at the sensor node, the FETPC at the controller node, and the networked data-dropout compensator (NDC) at the actuator node. For facilitating the theory development later, the following assumptions are given first.

*Assumption 1:* The sensor is clock driven and system states are sampled at a constant period. The controller and the actuator are event driven. The predictive control packets are transmitted carrying the time stamps. Then, by using the time-stamp technology, the NDC in Fig. 1 uses a data buffer to sequentially store the predictive control packets generated by FETPC. The delay caused by networks is not considered in this paper.

**Assumption 2:**  $\{t\}_{t=1}^{\infty}$  and  $\{t_k\}_{k=1}^{\infty}$  mean the set of sample instants and event-triggering instants at the sensor side, respectively. To include the data dropout,  $\{t_{s_i}\}_{i=1}^{\infty}$  is utilized to describe the set of event-triggering instants on which the triggering states  $\{x(t_{s_i})\}_{i=1}^{\infty}$  are successfully sent to the controller, and correspondingly the predictive control packets  $U_{t_{s_i}}$  are successfully sent to the actuator. Obviously,  $\{x(t_{s_i})\}_{i=1}^{\infty} \subseteq \{x(t_k)\}_{k=1}^{\infty} \subseteq \{x(t)\}_{t=1}^{\infty}$ .

**Assumption 3:** There is a communication network between the sensor and the controller, and the controller and the actuator are directly point-to-point connected without communication networks. The number of consecutive lost data occurring in the communication networks from the sensor to the controller is upper bounded by  $p$ .

Considering the communication networks with the limited bandwidth, the ETCS in event trigger 1 is introduced to alleviate the communication burden without damaging the stability of the studied system. Once the state  $x(t_k)$  at the triggering instants  $t_k$  ( $k = 1, 2, \dots$ ) is transmitted, the next triggering instant is up to the following condition [1], [28]:

$$t_{k+1} = t_k + \min\{M, r_k\}$$

$$r_k = \min_r \{r | \delta^T(t) \Phi \delta(t) > \sigma x^T(t) \Phi x(t)\} \quad (5)$$

where  $\delta(t) = x(t) - x(t_k)$  means the difference value of the current-sampled states  $x(t)$  and the latest transmitted states  $x(t_k)$ ,  $t = t_k + r$ ,  $r = 1, 2, \dots$ ; the positive integer  $M$  is the upper bound limiting the interval of adjacent triggering instants;  $\sigma$  is a given scalar parameter subject to  $0 < \sigma < 1$ ; and the positive-definite weighting matrix  $\Phi$  is symmetric and need to be found later.

**Remark 1:** Defining  $h(t, t_k) \triangleq \delta^T(t) \Phi \delta(t) - \sigma x^T(t) \Phi x(t)$ , it can be seen from ETCS (5) that the sampled states will not be transmitted over the communication networks unless  $r_k > M$  or  $h(t, t_k) > 0$  is satisfied. Therefore, the introduced ETCS (5) can significantly decrease the number of sampled states to be transmitted over the communication networks.

### B. FETPC Under Imperfect Premise Matching

Due to the unreliability of the communication networks, not all triggered packets can be sent over the networks successfully. In this section, the FETPC is constructed to compensate for the effect of data dropout in the transmissions.

Considering the communication networks between the sensor and the controller, the model of the FETPC cannot share the same premise variables as the studied fuzzy system due to the effect of the networks. The specific structure of the fuzzy controller has been delineated in Fig. 1, and the model of the FETPC is described as follows:

FETPC rule  $w$ : IF  $g_1(\hat{x}(t))$  is  $M_1^w$  and  $\dots$  and  $g_p(\hat{x}(t))$  is  $M_p^w$ , THEN

$$\hat{x}(t+1) = \hat{A}_w \hat{x}(t) + \hat{B}_w u(t) \quad (6)$$

where  $M_d^w$  ( $w = 1, \dots, \bar{r}$ ;  $d = 1, 2, \dots, p$ ) denote the fuzzy sets;  $g_d(\hat{x}(t))$  denote the premise variables;  $\bar{r}$  and  $p$  are, respectively, the number of fuzzy IF-THEN rules and the premise

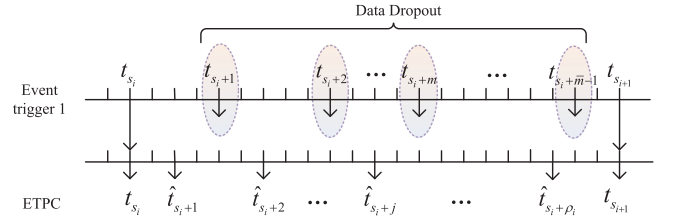


Fig. 2. Example of event triggering time evolution.

variables;  $\hat{x}(t) \in \mathbb{R}^n$  and  $u(t) \in \mathbb{R}^m$  represent the state vector and the control input vector of the model; and  $\hat{A}_w$  and  $\hat{B}_w$  represent constant matrices of the model with appropriate dimensions.

**Remark 2:** If  $A_v = \hat{A}_w$  and  $B_v = \hat{B}_w$ , it means the system matrices of the fuzzy plant can be obtained explicitly. If  $A_v = \hat{A}_w + \Delta A_{vw}$  and  $B_v = \hat{B}_w + \Delta B_{vw}$ , it takes the parameter uncertainties of the fuzzy plant into account.

The global dynamic model of the FETPC can be represented as

$$\hat{x}(t+1) = \sum_{w=1}^{\bar{r}} \eta_w(g(\hat{x})) [\hat{A}_w \hat{x}(t) + \hat{B}_w u(t)] \quad (7)$$

where

$$\eta_w(g(\hat{x})) = \frac{\beta_w(g(\hat{x}))}{\sum_{w=1}^{\bar{r}} \beta_w(g(\hat{x}))} \geq 0$$

$$\beta_w(g(\hat{x})) = \prod_{d=1}^p M_d^w(g_d(\hat{x})), \quad \sum_{w=1}^{\bar{r}} \eta_w(g(\hat{x})) = 1.$$

$M_d^w(g_d(\hat{x}))$  denotes the membership value of  $g_d(\hat{x})$  in  $M_d^w$ . In addition, the control law shares the same premise with the FETPC rule, which is designed as follows:

Controller rule  $l$ : IF  $g_1(\hat{x}(t))$  is  $M_1^l$  and  $\dots$  and  $g_p(\hat{x}(t))$  is  $M_p^l$ , THEN

$$\hat{u}(t) = K_l \hat{x}(t) \quad (8)$$

where  $K_l$  ( $l = 1, 2, \dots, \bar{r}$ ) are the FETPC gains to be designed. Then, the global fuzzy controller can be represented as

$$\hat{u}(t) = \sum_{l=1}^{\bar{r}} \eta_l(g(\hat{x})) K_l \hat{x}(t). \quad (9)$$

**Remark 3:** Based on Assumption 3 and the event-triggered scheme (5), the next successfully transmitted instant of  $t_{s_i}$  is  $t_{s_{i+1}}$ , if  $t_{s_{i+1}} - t_{s_i} = t_{s_i+m} - t_{s_i}$ , it follows that  $t_{s_{i+1}} - t_{s_i} = t_{s_i+m} - t_{s_i} \leq t_{s_i+p} - t_{s_i} = (t_{s_i+p} - t_{s_i+p-1}) + (t_{s_i+p-1} - t_{s_i+p-2}) + \dots + (t_{s_i+1} - t_{s_i}) \leq M * p$ . During the time interval  $[t_{s_i}, t_{s_{i+1}}]$ , it worth noting that corresponding to the event-triggering instants  $\{t_{s_i+m}\}_{m=1}^{s_{i+1}-s_i-1}$  of event trigger 1, the triggering states are lost because of the characteristics of the networks, as shown in Fig. 2. This fact will be used in the following section to get the ETPC signals.

### C. Model of Networked T-S Fuzzy Systems With ETPC

In this section, we will give the detailed design program for event-triggered predictor. At each time  $t_{s_i}$ ,  $i = 1, 2, \dots$ , once the triggering state  $x(t_{s_i})$  is successfully sent to the FETPC, based on model (7) and the control law (9), the states of the closed-loop system for future instants can be predicted as

$$\hat{x}(\hat{t}_{s_i+j} + k + 1 | t_{s_i}) = \sum_{w=1}^{\bar{r}} \eta_w(g(\hat{x})) \left[ \hat{A}_w \hat{x}(\hat{t}_{s_i+j} + k | t_{s_i}) + \hat{B}_w \sum_{l=1}^{\bar{r}} \eta_l(g(\hat{x})) K_l \hat{x}(\hat{t}_{s_i+j} | t_{s_i}) \right] \quad (10)$$

$$k \in \{0, 1, 2, \dots, \hat{t}_{s_i+j+1} - \hat{t}_{s_i+j} - 1\}, j \in \{0, 1, 2, \dots, \rho_i\}.$$

Notice that in (10), when  $j = 0$ , it satisfies that  $\hat{t}_{s_i+j} = t_{s_i}$  and  $\hat{x}(\hat{t}_{s_i+j} | t_{s_i}) = x(t_{s_i})$ ; when  $j = 0$  and  $k = 0$  simultaneously, it satisfies that  $\hat{x}(\hat{t}_{s_i+j} + k | t_{s_i}) = x(t_{s_i})$ . The closed-loop states deriving from (10) between  $t_{s_i}$  and  $t_{s_{i+1}}$  are  $x(t_{s_i}), \hat{x}(t_{s_i} + 1), \hat{x}(t_{s_i} + 2), \dots, \hat{x}(\hat{t}_{s_i+1}), \hat{x}(\hat{t}_{s_i+1} + 1), \hat{x}(\hat{t}_{s_i+1} + 2), \dots, \hat{x}(\hat{t}_{s_i+2}), \hat{x}(\hat{t}_{s_i+2} + 1), \hat{x}(\hat{t}_{s_i+2} + 2), \dots, \hat{x}(\hat{t}_{s_i+\rho_i-1}), \hat{x}(\hat{t}_{s_i+\rho_i-1} + 1), \hat{x}(\hat{t}_{s_i+\rho_i-1} + 2), \dots, \hat{x}(\hat{t}_{s_i+\rho_i})$ .

In (10), the predictive triggering instants  $\hat{t}_{s_i+j+1}$  next to the latest ones  $\hat{t}_{s_i+j}$  are determined by the ETCS

$$\begin{aligned} \hat{t}_{s_i+j+1} &= \hat{t}_{s_i+j} + \min\{M, r_{s_i+j}\} \\ r_{s_i+j} &= \min_r \{r | \hat{\delta}^T(t) \Phi \hat{\delta}(t) > \sigma \hat{x}^T(t | t_{s_i}) \Phi \hat{x}(t | t_{s_i})\} \quad (11) \\ j &\in \{0, 1, 2, \dots, \rho_i\} \end{aligned}$$

where  $\hat{\delta}(t) = \hat{x}(t | t_{s_i}) - \hat{x}(\hat{t}_{s_i+j} | t_{s_i})$ ,  $t = \hat{t}_{s_i+j} + r$ ,  $r = 1, 2, \dots$ , when  $j = 0$ ,  $\hat{t}_{s_i+j} = t_{s_i}$ , and  $\hat{x}(\hat{t}_{s_i+j} | t_{s_i}) = x(t_{s_i})$ ;  $\sigma$  and  $\Phi$  are the same as the parameters given in (5).  $\{\hat{t}_{s_i+j}\}_{j=1}^{\rho_i}$  ( $\hat{t}_{s_i+1} < \hat{t}_{s_i+2} < \dots < \hat{t}_{s_i+\rho_i}$ ) denotes the set of the predictive event-triggered instants and it satisfies  $\hat{t}_{s_i+\rho_i} \leq t_{s_i} + M * p < \hat{t}_{s_i+\rho_i+1}$  with  $\rho_i$ . An example of time evolution describing the triggering instants of event triggers 1 and 2 is shown in Fig. 2.

Corresponding to the above-mentioned predictive event-triggered instants in (11), the predictive states are  $x(t_{s_i}), \hat{x}(\hat{t}_{s_i+1}), \hat{x}(\hat{t}_{s_i+2}), \dots, \hat{x}(\hat{t}_{s_i+\rho_i})$ , and the predictive control signals are represented as

$$\begin{aligned} \hat{u}(\hat{t}_{s_i+j} | t_{s_i}) &= \sum_{l=1}^{\bar{r}} \eta_l(g(\hat{x})) K_l \hat{x}(\hat{t}_{s_i+j} | t_{s_i}) \quad (12) \\ j &\in \{0, 1, 2, \dots, \rho_i\}. \end{aligned}$$

Then, the control sequences in (12) are packaged into a single data packet as  $U_{t_{s_i}}$

$$U_{t_{s_i}} \triangleq [u(t_{s_i}), \hat{u}(\hat{t}_{s_i+1} | t_{s_i}), \hat{u}(\hat{t}_{s_i+2} | t_{s_i}), \dots, \hat{u}(\hat{t}_{s_i+\rho_i} | t_{s_i})] \quad (13)$$

and transmitted through the controller-to-actuator link.

**Remark 4:** If event trigger 2 is not introduced as the existing works in [19], [33], and [34], the control packet will be

$$\begin{aligned} \tilde{U}_{t_{s_i}} &\triangleq [u(t_{s_i}), \hat{u}(t_{s_i} + 1), \hat{u}(t_{s_i} + 2), \dots, \\ &\hat{u}(\hat{t}_{s_i+1} | t_{s_i}), \hat{u}(\hat{t}_{s_i+1} + 1 | t_{s_i}), \hat{u}(\hat{t}_{s_i+1} + 2 | t_{s_i}), \dots, \\ &\hat{u}(\hat{t}_{s_i+2} | t_{s_i}), \hat{u}(\hat{t}_{s_i+2} + 1 | t_{s_i}), \hat{u}(\hat{t}_{s_i+2} + 2 | t_{s_i}), \dots, \\ &\hat{u}(\hat{t}_{s_i+\rho_i} | t_{s_i})]. \end{aligned}$$

However, in this paper, since the introduction of event trigger 2, the instants to transmit the predicted values are determined by the ETCS (11); only the predictive states at triggering instants, i.e.,  $x(t_{s_i}), \hat{x}(\hat{t}_{s_i+1}), \hat{x}(\hat{t}_{s_i+2}), \dots, \hat{x}(\hat{t}_{s_i+\rho_i})$ , are used to get the control signals. As a result, the computer complexity and size of control packet  $U_{t_{s_i}}$  in (13) is much smaller.

From the above description, it is known that  $t_{s_{i+1}} \leq t_{s_i} + M * p$ , and it is noted that  $\hat{t}_{s_i+\rho_i} \leq t_{s_i} + M * p < \hat{t}_{s_i+\rho_i+1}$ ,  $i = 1, 2, \dots$ , so the plant can be well controlled during the time interval  $[t_{s_i}, t_{s_{i+1}})$ . For  $t \in [t_{s_i}, t_{s_{i+1}})$ , it is not until the buffer in NDC receives the control packet  $U_{t_{s_i}}$  that the former one  $U_{t_{s_{i-1}}}$  is discarded, and then the NDC will select the appropriate control signal  $u(t) = \hat{u}(\hat{t}_{s_i+j} | t_{s_i})$  in  $U_{t_{s_i}}$  for  $\hat{t}_{s_i+j} \leq t < \hat{t}_{s_i+j+1}$  as the input of actuator.  $U_{t_{s_i}}$  works until the next control packet  $U_{t_{s_{i+1}}}$  arrives during  $[t_{s_{i+1}}, t_{s_{i+2}})$ . Therefore, the data dropout induced by the communication networks can be actively compensated based on the method proposed in this paper.

Based on the description above, system (2) and the FETPC (7) under the event-triggering condition (5) and (11) can be formulated as the closed-loop system

$$\begin{aligned} x(t+1) &= \sum_{v=1}^{\bar{r}} \mu_v(x) [A_v x(t) + B_v \hat{u}(\hat{t}_{s_i+j} | t_{s_i})] \quad (14) \\ \hat{x}(t+1) &= \sum_{w=1}^{\bar{r}} \eta_w(\hat{x}) [\hat{A}_w \hat{x}(t) + \hat{B}_w \hat{u}(\hat{t}_{s_i+j} | t_{s_i})], t \in \Theta_{m,j}^i \quad (15) \end{aligned}$$

where  $\Theta_{m,j}^i \triangleq [t_{s_i}, t_{s_{i+1}}) \cap [t_{s_i+m}, t_{s_{i+m+1}}) \cap [\hat{t}_{s_i+j}, \hat{t}_{s_i+j+1})$ , and the interval  $[t_{s_i}, t_{s_{i+1}})$  can be divided into some subintervals as  $[t_{s_i}, t_{s_{i+1}}) = \cup_{m=0}^{s_{i+1}-s_i-1} \cup_{j=0}^{\rho_i} \Theta_{m,j}^i$ .

Then, defining the error vectors  $\hat{\delta}_{ij}(t) = \hat{x}(t) - \hat{x}(\hat{t}_{s_i+j} | t_{s_i})$ ,  $\delta_{im}(t) = x(t) - x(t_{s_i+m})$ , and  $e(t) \triangleq x(t) - \hat{x}(t)$ , and combining (12), system (14)–(15) can be written as

$$\begin{aligned} x(t+1) &= \sum_{v=1}^{\bar{r}} \sum_{l=1}^{\bar{r}} \mu_v(x) \eta_l(\hat{x}) [F_1(t)] \quad (16) \\ e(t+1) &= \sum_{v=1}^{\bar{r}} \sum_{w=1}^{\bar{r}} \sum_{l=1}^{\bar{r}} \mu_v(x) \eta_w(\hat{x}) \eta_l(\hat{x}) [F_2(t)], t \in \Theta_{m,j}^i \quad (17) \end{aligned}$$

where  $F_1(t) = (A_v + B_v K_l)x(t) - B_v K_l e(t) - B_v K_l \hat{\delta}_{ij}(t)$ ,  $F_2(t) = (A_v - \hat{A}_w + (B_v - \hat{B}_w) K_l)x(t) + (\hat{A}_w - (B_v - \hat{B}_w) K_l)e(t) - (B_v - \hat{B}_w) K_l \hat{\delta}_{ij}(t)$ .

Furthermore, the state vector is augmented as  $\varsigma = [x^T(t), e^T(t)]^T$ , and the system (16)–(17) can be uniformly represented as

$$\varsigma(t+1) = \sum_{v=1}^{\bar{r}} \sum_{w=1}^{\bar{r}} \sum_{l=1}^{\bar{r}} \mu_v(x) \eta_w(\hat{x}) \eta_l(\hat{x}) [\chi_{vwl}] \quad (18)$$

where

$$\chi_{vwl} = \Pi_{vwl} \varsigma(t) + \Xi_{vwl} \hat{\delta}_{ij}(t)$$

with

$$\Pi_{vwl} = \begin{bmatrix} A_v + B_v K_l & -B_v K_l \\ A_v - \hat{A}_w + (B_v - \hat{B}_w) K_l & \hat{A}_w - (B_v - \hat{B}_w) K_l \end{bmatrix}$$

$$\Xi_{vwl} = \begin{bmatrix} -B_v K_l \\ -(B_v - \hat{B}_w) K_l \end{bmatrix}.$$

*Remark 5:* According to the event-triggering condition (5) as well as (11), no event is triggered at time  $t \in (t_{s_i+m}, t_{s_i+m+1})$  in event trigger 1 or at time  $t \in (\hat{t}_{s_i+j}, \hat{t}_{s_i+j+1})$  in even trigger 2; thus, it can be concluded that

$$\hat{\delta}_{ij}^T(t) \Phi \hat{\delta}_{ij}(t) \leq \sigma \hat{x}^T(t) \Phi \hat{x}(t) \quad (19)$$

$$\delta_{im}^T(t) \Phi \delta_{im}(t) \leq \sigma x^T(t) \Phi x(t). \quad (20)$$

The Lemma introduced from [30] will be essential in establishing the subsequent results.

*Lemma 1:* [30] Suppose that matrices  $\{\Psi_i\}_{i=1}^s \in \mathbb{R}^{n \times m}$  and a positive-semidefinite matrix  $P \in \mathbb{R}^{n \times n}$  are given. If  $\sum_{i=1}^s \nu_i = 1$  and  $0 \leq \nu_i \leq 1$ , then

$$\left( \sum_{i=1}^s \nu_i \Psi_i \right)^T P \left( \sum_{i=1}^s \nu_i \Psi_i \right) \leq \sum_{i=1}^s \nu_i \Psi_i^T P \Psi_i.$$

### III. MAIN RESULTS

In this section, a stability criterion and two FETPC design criteria for the system (18) with plant uncertainties will be derived by the use of a common Lyapunov functional and non-PDC fuzzy control design approaches.

The stability criterion for the system (18) is presented as follows.

*Theorem 1:* Under Assumptions 1–3, for given matrices  $K_l$ , parameter  $0 < \sigma < 1$ , and membership functions satisfying  $\eta_l(\hat{x}) - \rho_l \mu_l(x) \geq 0$  ( $0 < \rho_l < 1$ ), the system (18) is asymptotically stable under the ETPC scheme in this paper if there exist matrices  $Q > 0$ ,  $\Phi > 0$  and arbitrary matrices  $\Lambda_v$ ,  $\Lambda_l$  of appropriate dimensions, with the result that the following linear matrix inequalities (LMIs) hold with  $v, w, l = 1, \dots, \bar{r}$ :

$$\Upsilon_{vwl} - \Lambda_v < 0 \quad (21)$$

$$\rho_v \Upsilon_{vww} - \rho_v \Lambda_v + \Lambda_v < 0 \quad (22)$$

$$\rho_l \Upsilon_{vwl} + \rho_v \Upsilon_{lww} - \rho_l \Lambda_v - \rho_v \Lambda_l + \Lambda_v + \Lambda_l < 0 \quad (23)$$

where

$$\Upsilon_{vwl} = \begin{bmatrix} -Q + \sigma \Omega & * & * & * \\ 0 & -\Phi & * & * \\ 0 & 0 & -\Phi & * \\ Q \Pi_{vwl} & Q \Xi_{vwl} & 0 & -Q \end{bmatrix}.$$

*Proof:* A Lyapunov functional is chosen as

$$V(\varsigma(t), t) = \varsigma^T(t) Q \varsigma(t) \quad (24)$$

where  $Q > 0$  is a symmetric matrix. During  $t \in \Theta_{mj}^i$ , taking the inequalities (19) and (20) into account, the difference of  $V(\varsigma(t), t)$  yields

$$\begin{aligned} \Delta V(\varsigma(t), t) &= V(\varsigma(t+1), t+1) - V(\varsigma(t), t) \\ &\leq \varsigma^T(t+1) Q \varsigma(t+1) - \varsigma^T(t) Q \varsigma(t) \\ &\quad + \sigma [x(t) - e(t)]^T \Phi [x(t) - e(t)] - \hat{\delta}_{ij}^T(t) \Phi \hat{\delta}_{ij}(t) \\ &\quad + \sigma x^T(t) \Phi x(t) - \delta_{im}^T(t) \Phi \delta_{im}(t) \\ &\quad + \sigma e^T(t) \Phi e(t). \end{aligned} \quad (25)$$

By using Lemma 1, one has

$$\begin{aligned} &\varsigma^T(t+1) Q \varsigma(t+1) \\ &= \left\{ \sum_{v=1}^{\bar{r}} \sum_{w=1}^{\bar{r}} \sum_{l=1}^{\bar{r}} \mu_v(x) \eta_w(\hat{x}) \eta_l(\hat{x}) [\Pi_{vwl} \varsigma(t) + \Xi_{vwl} \hat{\delta}_{ij}(t)] \right\}^T \\ &\quad Q \left\{ \sum_{v=1}^{\bar{r}} \sum_{w=1}^{\bar{r}} \sum_{l=1}^{\bar{r}} \mu_v(x) \eta_w(\hat{x}) \eta_l(\hat{x}) [\Pi_{vwl} \varsigma(t) + \Xi_{vwl} \hat{\delta}_{ij}(t)] \right\} \\ &\leq \sum_{v=1}^{\bar{r}} \mu_v(x) \left\{ \sum_{w=1}^{\bar{r}} \sum_{l=1}^{\bar{r}} \eta_w(\hat{x}) \eta_l(\hat{x}) [\Pi_{vwl} \varsigma(t) + \Xi_{vwl} \hat{\delta}_{ij}(t)] \right\}^T \\ &\quad Q \left\{ \sum_{w=1}^{\bar{r}} \sum_{l=1}^{\bar{r}} \eta_w(\hat{x}) \eta_l(\hat{x}) [\Pi_{vwl} \varsigma(t) + \Xi_{vwl} \hat{\delta}_{ij}(t)] \right\} \\ &\leq \sum_{v=1}^{\bar{r}} \sum_{w=1}^{\bar{r}} \mu_v(x) \eta_w(\hat{x}) \left\{ \sum_{l=1}^{\bar{r}} \eta_l(\hat{x}) [\Pi_{vwl} \varsigma(t) + \Xi_{vwl} \hat{\delta}_{ij}(t)] \right\}^T \\ &\quad Q \left\{ \sum_{l=1}^{\bar{r}} \eta_l(\hat{x}) [\Pi_{vwl} \varsigma(t) + \Xi_{vwl} \hat{\delta}_{ij}(t)] \right\} \\ &\leq \sum_{v=1}^{\bar{r}} \sum_{w=1}^{\bar{r}} \sum_{l=1}^{\bar{r}} \mu_v(x) \eta_w(\hat{x}) \eta_l(\hat{x}) \left[ \Pi_{vwl} \varsigma(t) + \Xi_{vwl} \hat{\delta}_{ij}(t) \right]^T \\ &\quad Q \left[ \Pi_{vwl} \varsigma(t) + \Xi_{vwl} \hat{\delta}_{ij}(t) \right]. \end{aligned} \quad (26)$$

Considering (25) and (26) leads to

$$\Delta V(\varsigma(t), t) \leq \sum_{v=1}^{\bar{r}} \sum_{w=1}^{\bar{r}} \sum_{l=1}^{\bar{r}} \mu_v(x) \eta_w(\hat{x}) \eta_l(\hat{x}) [F_3(t)] \quad (27)$$

where  $F_3(t) = [\Pi_{vwl} \varsigma(t) + \Xi_{vwl} \hat{\delta}_{ij}(t)]^T Q [\Pi_{vwl} \varsigma(t) + \Xi_{vwl} \hat{\delta}_{ij}(t)] - \varsigma^T(t) Q \varsigma(t) - \hat{\delta}_{ij}^T(t) \Phi \hat{\delta}_{ij}(t) - \delta_{im}^T(t) \Phi \delta_{im}(t) + \sigma \varsigma^T(t) \Omega \varsigma(t)$ , and  $\Omega = \begin{bmatrix} 2\Phi & -\Phi \\ -\Phi & 2\Phi \end{bmatrix}$ .

Define

$$\xi^T(t) = \begin{bmatrix} \varsigma^T(t) & \delta_{ij}^T(t) & \delta_{im}^T(t) \end{bmatrix}.$$

It follows from (27) that

$$\Delta V(\varsigma(t), t) = \sum_{v=1}^{\bar{r}} \sum_{w=1}^{\bar{r}} \sum_{l=1}^{\bar{r}} \mu_v(x) \eta_w(\hat{x}) \eta_l(\hat{x}) \xi^T(t) \Upsilon_{vwl} \xi(t). \quad (28)$$

Consider

$$\begin{aligned} & \sum_{v=1}^{\bar{r}} \sum_{w=1}^{\bar{r}} \sum_{l=1}^{\bar{r}} \mu_v(x) \eta_w(\hat{x}) [\mu_l(x) - \eta_l(\hat{x})] \Lambda_v \\ &= \sum_{v=1}^{\bar{r}} \sum_{w=1}^{\bar{r}} \mu_v(x) \eta_w(\hat{x}) \left( \sum_{l=1}^{\bar{r}} \mu_l(x) - \sum_{l=1}^{\bar{r}} \eta_l(\hat{x}) \right) \Lambda_v \\ &= \sum_{v=1}^{\bar{r}} \sum_{w=1}^{\bar{r}} \mu_v(x) \eta_w(\hat{x}) (1 - 1) \Lambda_v = 0 \end{aligned}$$

in [26], where  $\Lambda_v = \Lambda_v^T \in \mathbb{R}^{6n \times 6n}$ ,  $v = 1, 2, \dots, \bar{r}$ , are arbitrary matrices. Introducing these terms to (28) yields

$$\begin{aligned} & \sum_{v=1}^{\bar{r}} \sum_{w=1}^{\bar{r}} \sum_{l=1}^{\bar{r}} \mu_v(x) \eta_w(\hat{x}) \eta_l(\hat{x}) \Upsilon_{vwl} \\ &\leq \sum_{v=1}^{\bar{r}} \sum_{w=1}^{\bar{r}} \mu_v^2(x) \eta_w(\hat{x}) (\rho_v \Upsilon_{vww} - \rho_v \Lambda_v + \Lambda_v) \\ &\quad + \sum_{v=1}^{\bar{r}} \sum_{w=1}^{\bar{r}} \sum_{l=1}^{\bar{r}} \mu_v(x) \eta_w(\hat{x}) (\eta_l(\hat{x}) - \rho_l \mu_l(x)) (\Upsilon_{vwl} - \Lambda_v) \\ &\quad + \sum_{v=1}^{\bar{r}} \sum_{w=1}^{\bar{r}} \sum_{l < v}^{\bar{r}} \mu_v(x) \eta_w(\hat{x}) \mu_l(x) \Gamma_{vwl} \end{aligned}$$

where  $\Gamma_{vwl} = \rho_l \Upsilon_{vwl} + \rho_v \Upsilon_{lww} - \rho_l \Lambda_v - \rho_v \Lambda_l + \Lambda_v + \Lambda_l$  with  $\eta_l(\hat{x}) - \rho_l \mu_l(x) \geq 0$  for all  $l$ . Then, let (21)–(23) hold for all  $v, w, l = 1, 2, \dots, \bar{r}$ ; the following can be derived:

$$\Delta V(\varsigma(t), t) < 0. \quad (29)$$

Then, a scalar  $\epsilon > 0$  is existent and satisfies  $\Delta V(\varsigma(t), t) \leq -\epsilon \|\xi^2(t)\|$  for  $\xi(t) \neq 0$ ; thus,  $\Delta V(\varsigma(t), t) < 0$ , and the asymptotical stability of system (18) is ensured. ■

**Remark 6:** Under the proposed framework of networked T-S fuzzy system, the system (2) is controlled by the imperfect premise matching FETPC (7), thus in (18)  $\mu_v(x) \neq \eta_l(\hat{x})$ . If  $\mu_v(x) = \eta_l(\hat{x})$ , which means the fuzzy controller has the same premise membership functions as those of the T-S fuzzy model, the conventional PDC technology [8] can be used to derive the corresponding results. Thus, the PDC technology can be regarded as a special case of the method proposed in this paper, which means that the proposed fuzzy controller fuses the superiorities of both perfect and imperfect premise matching.

Based on Theorem 1, the following theorem gives a method to find the event-triggering parameter and the FETPC gains simultaneously.

**Theorem 2:** Under Assumptions 1–3, for given matrices  $\hat{A}_w$ ,  $\hat{B}_w$ , parameter  $0 < \sigma < 1$ , and membership functions satisfying  $\eta_l(\hat{x}) - \rho_l \mu_l(x) \geq 0$  ( $0 < \rho_l < 1$ ), the system (18) will be asymptotically stable under the ETPC scheme in this paper if there exist matrices  $\bar{X} > 0$ ,  $\bar{\Phi} > 0$ ,  $Y_l$  and arbitrary matrices  $\bar{\Lambda}_v$ ,  $\bar{\Lambda}_l$  of appropriate dimensions, with the result that the following LMIs hold for all  $v, w, l = 1, \dots, \bar{r}$ :

$$\bar{\Upsilon}_{vwl} - \bar{\Lambda}_v < 0 \quad (30)$$

$$\rho_v \bar{\Upsilon}_{vww} - \rho_v \bar{\Lambda}_v + \bar{\Lambda}_v < 0 \quad (31)$$

$$\rho_l \bar{\Upsilon}_{vwl} + \rho_v \bar{\Upsilon}_{lww} - \rho_l \bar{\Lambda}_v - \rho_v \bar{\Lambda}_l + \bar{\Lambda}_v + \bar{\Lambda}_l < 0 \quad (32)$$

where

$$\bar{\Upsilon}_{vwl} = \begin{bmatrix} \bar{\Upsilon}_{11} & \bar{\Upsilon}_{12} & * & * & * \\ 0 & 0 & -\bar{\Phi} & * & * \\ 0 & 0 & 0 & -\bar{\Phi} & * \\ \bar{\Upsilon}_{41} & \bar{\Upsilon}_{42} & \bar{\Upsilon}_{43} & 0 & \bar{\Upsilon}_{45} \end{bmatrix}$$

with

$$\bar{\Upsilon}_{11} = \begin{bmatrix} 2\sigma\bar{\Phi} - \bar{X} \\ -\sigma\bar{\Phi} \end{bmatrix}$$

$$\bar{\Upsilon}_{12} = \begin{bmatrix} * \\ 2\sigma\bar{\Phi} - \bar{X} \end{bmatrix}$$

$$\bar{\Upsilon}_{41} = \begin{bmatrix} A_v \bar{X} + B_v Y_l \\ (A_v - \hat{A}_w) \bar{X} + (B_v - \hat{B}_w) Y_l \end{bmatrix}$$

$$\bar{\Upsilon}_{42} = \begin{bmatrix} -B_v Y_l \\ \hat{A}_w \bar{X} - (B_v - \hat{B}_w) Y_l \end{bmatrix}$$

$$\bar{\Upsilon}_{43} = \begin{bmatrix} -B_v Y_l \\ -(B_v - \hat{B}_w) Y_l \end{bmatrix}$$

$$\bar{\Upsilon}_{45} = \begin{bmatrix} -\bar{X} & * \\ 0 & -\bar{X} \end{bmatrix}.$$

Thus, the parameter  $\bar{\Phi}$  of ETCS (5) and (11) and the fuzzy controller gains  $K_l$  are obtained by

$$\bar{\Phi} = \bar{X}^{-1} \bar{\Phi} \bar{X}^{-1}, K_l = Y_l \bar{X}^{-1}. \quad (33)$$

**Proof:** Set  $Q = \text{diag}\{\bar{Q}, \bar{Q}\}$  and define  $\bar{X} = \bar{Q}^{-1}$ ,  $X = \text{diag}\{\bar{X}, \bar{X}\}$ ,  $\bar{X} \bar{\Phi} \bar{X} = \bar{\Phi}$ , and  $Y_l = K_l \bar{X}$ . Pre- and postmultiplying (21)–(23) by  $\text{diag}\{X, \bar{X}, \bar{X}, X\}$ , (36)–(38) can be obtained. ■

In what follows, the uncertainties of the fuzzy plant are taken into account. Without loss of generality, the uncertainties are given as

$$\begin{aligned} [\Delta A_{vw}(t) \quad \Delta B_{vw}(t)] &= [A_v - \hat{A}_w \quad B_v - \hat{B}_w] \\ &= D_{vw} \mathfrak{F}_{vw}(t) [E_{vw}^A \quad E_{vw}^B] \end{aligned} \quad (34)$$

where  $\Delta A_{vw}(t)$  and  $\Delta B_{vw}(t)$  are the time-varying matrices with appropriate dimensions;  $D_{vw}$  and  $E_{vw}^A, E_{vw}^B$ ,  $v = 1, \dots, \bar{r}$ ,  $w = 1, \dots, \bar{r}$ , are known constant real matrices with appropriate dimensions; and  $\mathfrak{F}_{vw}(t)$  are unknown real time-varying matrices with Lebesgue measurable elements bounded by

$$\mathfrak{F}_{vw}^T(t) \mathfrak{F}_{vw}(t) \leq I. \quad (35)$$

The following theorem can be derived.

**Theorem 3:** For given matrices  $\hat{A}_w, \hat{B}_w, D_{vw}, E_{vw}^A, E_{vw}^B$ , parameter  $0 < \sigma < 1$ , and membership functions satisfying  $\eta_l(\hat{x}) - \rho_l \mu_l(x) \geq 0$  ( $0 < \rho_l < 1$ ), the system (18) will be asymptotically stable under the ETPC scheme in this paper if there exist matrices  $\bar{X} > 0, \bar{\Phi} > 0, Y_l$  and arbitrary matrices  $\bar{A}_v, \bar{A}_l^T$  of appropriate dimensions, with the result that the following LMIs hold for all  $v, w, l = 1, \dots, \bar{r}$ :

$$\bar{\Upsilon}'_{vwl} - \bar{A}_v' < 0 \quad (36)$$

$$\rho_v \bar{\Upsilon}'_{vww} - \rho_v \bar{A}_v' + \bar{A}_v' < 0 \quad (37)$$

$$\rho_l \bar{\Upsilon}'_{vwl} + \rho_v \bar{\Upsilon}'_{lvw} - \rho_l \bar{A}_v' - \rho_v \bar{A}_l' + \bar{A}_v' + \bar{A}_l' < 0 \quad (38)$$

where

$$\bar{\Upsilon}'_{vwl} = \begin{bmatrix} \bar{\Upsilon}'_{11} & \bar{\Upsilon}'_{12} & * & * & * & * \\ 0 & 0 & -\bar{\Phi} & * & * & * \\ 0 & 0 & 0 & -\bar{\Phi} & * & * \\ \bar{\Upsilon}'_{41} & \bar{\Upsilon}'_{42} & \bar{\Upsilon}'_{43} & 0 & \bar{\Upsilon}'_{45} & * \\ \bar{\Upsilon}'_{51} & \bar{\Upsilon}'_{52} & \bar{\Upsilon}'_{53} & 0 & 0 & \bar{\Upsilon}'_{56} \end{bmatrix}$$

with

$$\bar{\Upsilon}'_{11} = \begin{bmatrix} 2\sigma\bar{\Phi} - \bar{X} \\ -\sigma\bar{\Phi} \end{bmatrix}, \bar{\Upsilon}'_{12} = \begin{bmatrix} * \\ 2\sigma\bar{\Phi} - \bar{X} \end{bmatrix}$$

$$\bar{\Upsilon}'_{41} = \begin{bmatrix} A_v \bar{X} + B_v Y_l \\ 0 \end{bmatrix}, \bar{\Upsilon}'_{42} = \begin{bmatrix} -B_v Y_l \\ \hat{A}_w \bar{X} \end{bmatrix}$$

$$\bar{\Upsilon}'_{43} = \begin{bmatrix} -B_v Y_l \\ 0 \end{bmatrix}, \bar{\Upsilon}'_{45} = \begin{bmatrix} -\bar{X} & * \\ 0 & -\bar{X} + (\varepsilon_1 + \varepsilon_2) D_{vw} D_{vw}^T \end{bmatrix}$$

$$\bar{\Upsilon}'_{51} = \begin{bmatrix} E_{vw}^A \bar{X} \\ E_{vw}^B Y_l \end{bmatrix}, \bar{\Upsilon}'_{52} = \bar{\Upsilon}'_{53} = \begin{bmatrix} 0 \\ -E_{vw}^B Y_l \end{bmatrix}$$

$$\bar{\Upsilon}'_{56} = \begin{bmatrix} -\varepsilon_1 I & * \\ 0 & -\varepsilon_2 I \end{bmatrix}.$$

Then, the parameter  $\Phi$  of ETCS (5) and (11) and the fuzzy controller gains  $K_l$  can be obtained by (33).

*Proof:* Based on the uncertainties described above,  $\bar{\Upsilon}_{vwl}$  in Theorem 2 can be written as

$$\bar{\Upsilon}_{vwl} = \bar{\Upsilon}_{vwl}^1 + \bar{h}_0 \mathfrak{F}_{vw}(t) \bar{h}_1 + \bar{h}_1^T \mathfrak{F}_{vw}^T(t) \bar{h}_0^T \quad (39)$$

$$+ \bar{h}_0 \mathfrak{F}_{vw}(t) \bar{h}_2 + \bar{h}_2^T \mathfrak{F}_{vw}^T(t) \bar{h}_0^T \quad (40)$$

where

$$\bar{\Upsilon}_{vwl}^1 = \begin{bmatrix} 2\sigma\bar{\Phi} - \bar{X} & * & * & * & * & * \\ -\sigma\bar{\Phi} & 2\sigma\bar{\Phi} - \bar{X} & * & * & * & * \\ 0 & 0 & -\bar{\Phi} & * & * & * \\ 0 & 0 & 0 & -\bar{\Phi} & * & * \\ A_v \bar{X} + B_v Y_l & -B_v Y_l & -B_v Y_l & 0 & -\bar{X} & * \\ 0 & \hat{A}_w \bar{X} & 0 & 0 & 0 & -\bar{X} \end{bmatrix}$$

$$\bar{h}_0 = [0 \ 0 \ 0 \ 0 \ 0 \ D_{vw}^T]^T$$

$$\bar{h}_1 = [E_{vw}^A \bar{X} \ 0 \ 0 \ 0 \ 0 \ 0]$$

$$\bar{h}_2 = [E_{vw}^B Y_l \ -E_{vw}^B Y_l \ -E_{vw}^B Y_l \ 0 \ 0 \ 0].$$

It is seen that there exist  $\varepsilon_1$  and  $\varepsilon_2$ , such that

$$\bar{h}_0 \mathfrak{F}_{vw}(t) \bar{h}_1 + \bar{h}_1^T \mathfrak{F}_{vw}^T(t) \bar{h}_0^T \leq \varepsilon_1 \bar{h}_0 \bar{h}_0^T + \varepsilon_1^{-1} \bar{h}_1^T \bar{h}_1$$

$$\bar{h}_0 \mathfrak{F}_{vw}(t) \bar{h}_2 + \bar{h}_2^T \mathfrak{F}_{vw}^T(t) \bar{h}_0^T \leq \varepsilon_2 \bar{h}_0 \bar{h}_0^T + \varepsilon_2^{-1} \bar{h}_2^T \bar{h}_2$$

for all  $\mathfrak{F}_{vw}(t)$  satisfying (35). Thus, it yields from (39) that

$$\bar{\Upsilon}_{vwl} \leq \bar{\Upsilon}_{vwl}^1 + (\varepsilon_1 + \varepsilon_2) \bar{h}_0 \bar{h}_0^T + \varepsilon_1^{-1} \bar{h}_1^T \bar{h}_1 + \varepsilon_2^{-1} \bar{h}_2^T \bar{h}_2. \quad (41)$$

Connecting with the Schur complement,  $\bar{\Upsilon}'_{vwl}$  in Theorem 3 can be obtained from (41).

#### IV. NUMERICAL EXAMPLES

To illustrate the validity of the ETPC scheme proposed throughout the paper for networked T-S fuzzy systems, the nonlinear mass-spring system [32] is used in this section, which is given as follows:

$$\dot{\varrho}_1 = \varrho_2 \quad (42)$$

$$\dot{\varrho}_2 = -0.01\varrho_1 - 0.67\varrho_1^3 + u \quad (43)$$

where  $\varrho_1 \in [-1, 1]$ . If the continuous T-S fuzzy model in [16] is discretized with  $h = 0.1$  s, it can be modeled by the discrete T-S fuzzy system as follows:

$$x(k+1) = \sum_{i=1}^2 \mu(f(x)) [A_i x(k) + B_i u(k)] \quad (44)$$

where

$$A_1 = \begin{bmatrix} 1.0000 & 0.1000 \\ -0.0010 & 1.0000 \end{bmatrix}, \quad B_1 = \begin{bmatrix} 0.0050 \\ 0.1000 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 0.9966 & 0.0999 \\ -0.0679 & 0.9966 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 0.0050 \\ 0.0999 \end{bmatrix}.$$

The fuzzy membership functions are chosen as

$$\mu_1(f(x)) = \frac{1}{1 + \exp(-2x_1(k))}, \quad \mu_2(f(x)) = 1 - \mu_1(f(x))$$

and the membership functions of the FETPC are

$$\eta_1(g(\hat{x})) = 0.99 \exp\left(\frac{-\hat{x}_1^2(k)}{2 \times 1.5^2}\right), \quad \eta_2(g(\hat{x})) = 1 - \eta_1(g(\hat{x})).$$

*Case 1:* Suppose  $\hat{A} = 1.01 * A$  and  $\hat{B} = 0.95 * B$ . With the given parameters  $\sigma = 0.2$  and  $\alpha = 0.1$ ,  $\rho_1 = 0.8$ ,  $\rho_2 = 0.95$ , which ensure that  $\eta_l(\hat{x}) - \rho_l \mu_l(x) \geq 0$  in Theorem 2, by using the LMI toolbox in MATLAB, the matrix  $\Phi$  in ETCS (5) and (11) and the corresponding FETPC gains  $K_1$  and  $K_2$  can be obtained as follows:

$$\Phi = \begin{bmatrix} 1.0662 & 0.2482 \\ 0.2483 & 0.9310 \end{bmatrix}$$

$$K_1 = [-0.2170 \quad -0.4024]$$

$$K_2 = [0.0672 \quad -0.1402].$$

With the triggering matrix  $\Phi$  and the feedback gain matrices  $K_1$  and  $K_2$  obtained above, initializing the state as  $x_0 = [0.5, -0.5]^T$  and setting the sampling period as  $h = 0.1$  s,

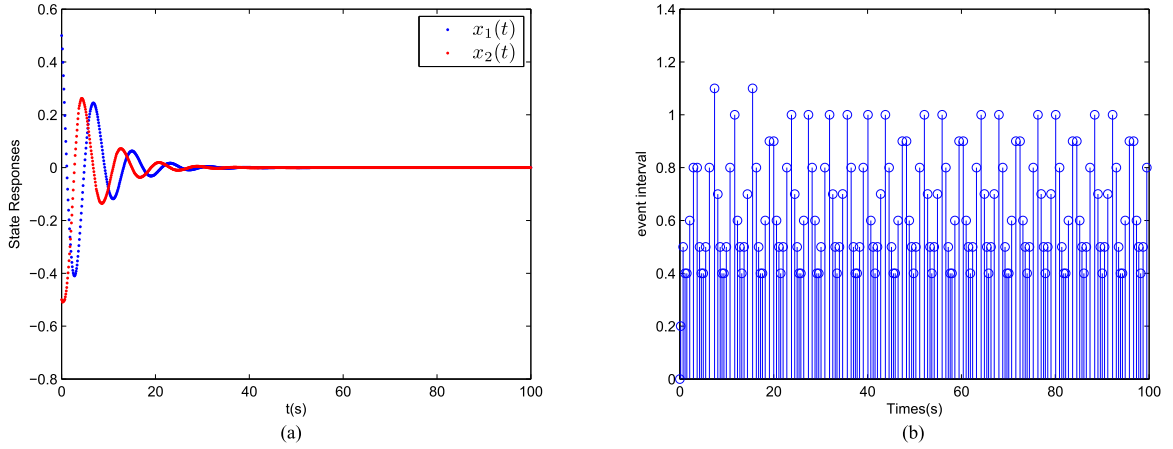


Fig. 3. Subcase 1-1 with data-dropout compensation. (a) State trajectories. (b) Event-triggered intervals.

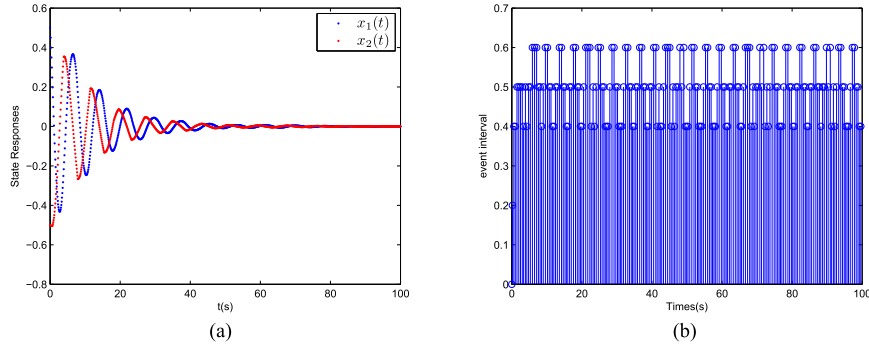


Fig. 4. Subcase 1-1 without data-dropout compensation. (a) State trajectories. (b) Event-triggered intervals.

we use the ETPC scheme raised in this paper to compensate for the data dropout. To show the advantages of the proposed imperfect premise matching FETPC, two cases are considered in the following: in subcase 1-1 the data dropout is regular and in subcase 1-2 the data dropout is random. The simulation time is set as  $T = 100$  s.

*Subcase 1-1:* If  $k$  is an event number, then each triggering state  $x(t_k)$ ,  $k = 1, 2, 3, \dots$ , sent from the sensor to the controller is assumed to be dropped out.

Fig. 3 depicts the state trajectories and the event-triggered intervals in the sensor node of the nonlinear mass-spring system with data-dropout compensation, respectively. One can see from Fig. 3 that with the proposed FETPC, the closed-loop system can be stabilized well and the communication burden can be considerably alleviated. Fig. 4 presents the state trajectories of the plant and the event-triggered intervals in the sensor node as has been done in [1] that the data dropout is not compensated but event trigger 1 is still existent.

It can be found that the system performance without data-dropout compensation is worse compared with that of the system with data-dropout compensation, even though relatively more triggering sampled states are needed.

*Subcase 1-2:* It is a more common that the data dropout is occurring stochastically. The random data-dropout instants and intervals are shown in Fig. 5(a).

By using the proposed method to compensate for the data dropout described in Fig. 5(a), the state trajectories and the

event-triggered intervals of the nonlinear mass-spring system are shown in Fig. 5(b) and (c), from which we can see that the system tends to be stable within about 30 s. On the other hand, if the data dropout is not compensated, similar to the conclusion derived in subcase 1-1, the state responses of the nonlinear mass-spring system are much worse than those of the system with data-dropout compensation, and it needs more time and more triggering sampled states to stabilize the studied system.

*Case 2:* Suppose  $\hat{A} = A$  and  $\hat{B} = B$ . With the given parameters given as in case 1, the matrix  $\Phi$  in ETCS (5) and (11) and the corresponding FETPC gains  $K_1$  and  $K_2$  can be obtained as follows:

$$\Phi = \begin{bmatrix} 0.7148 & 0.1143 \\ 0.1143 & 0.8117 \end{bmatrix}$$

$$K_1 = \begin{bmatrix} -0.1999 & -0.3584 \end{bmatrix}$$

$$K_2 = \begin{bmatrix} 0.0829 & -0.1196 \end{bmatrix}.$$

Since the data dropout in subcase 1-2 is occurring stochastically, we cannot compare cases 2 and 1 with the same data dropout. Making sure the data dropout the same as that of subcase 1-1, Fig. 6 depicts the state trajectories and the event-triggered intervals in event trigger 1 of the system with data-dropout compensation, respectively. From Fig. 6(a), we can find that the convergence rate is faster than that of Fig. 3(a). Thus, the uncertainties of the fuzzy plant will slow down the progress of stabilization.

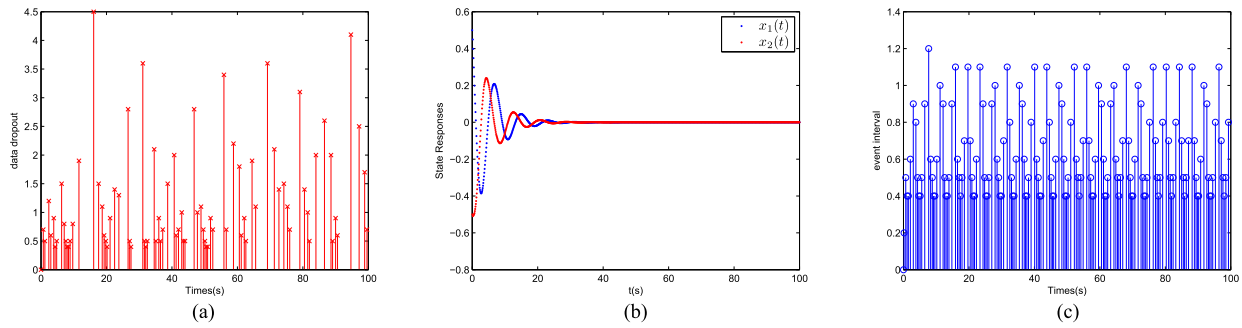


Fig. 5. Subcase 1-2 with data-dropout compensation. (a) Data-dropout instants and intervals. (b) State trajectories. (c) Event-triggered intervals.

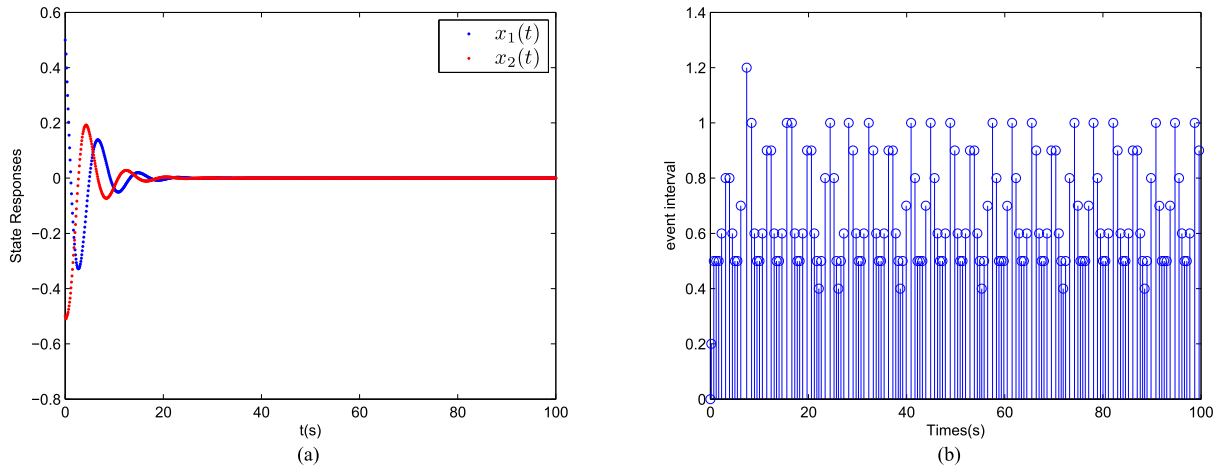


Fig. 6. Case 2 with data-dropout compensation. (a) State trajectories. (b) Event-triggered intervals.

## V. CONCLUSION

An FETPC design method for networked T-S fuzzy system has been presented with imperfect premise matching. The event triggers have been employed to alleviate the communication burden while keeping the desired control performance. The fuzzy predictive controller has been well constructed to predict the lost data between two successfully transmitted instants. A stability criterion and two stabilization criteria have been presented to ensure the stability of the studied system based on the common Lyapunov function method. Moreover, the uncertainties of the fuzzy plant are also taken into account. The nonlinear mass-spring system has been used to demonstrate the validity of the proposed approach. As a future work, we will study the FETPC for networked nonlinear system under consideration of the communication delay and data dropout together, and similar to [35] and [36], the piecewise Lyapunov functional methods will be used to obtain less conservative results.

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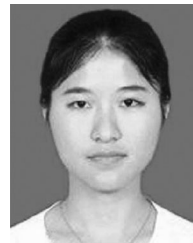
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